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SUPERSEDING
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23 December 1965

# MILITARY SPECIFICATION

# TETRAETHYLENEPENTAMINE

This specification is approved for use by all departments and agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers one grade of tetraethylenepentamine.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116 Preservation - Packaging, Methods of

MIL-P-14232 Parts, Equipment, and Tools for Army Materiel, Packaging and Packing of

STANDARDS

Military

MIL-STD-129 Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

FSC 6810

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

| | |
|---|---|
| ASTM D 86 | Distillation of Gasoline, Naphtha, Kerosene, and Similar Petroleum Products. |

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Uniform Classification Committee

Uniform Freight Classification Rules

(Application for copies should be addressed to Uniform Classification Committee, 202 Chicago Union Station, Chicago, Illinois, 60606.)

American Trucking Association Publication

National Motor Freight Classification Rules and Container Regulations

(Application for copies should be addressed to the National Classification Board, 1424 Sixteenth Street, N.W., Washington D.C. 20013.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), a preproduction sample shall meet the requirements of the specification.

3.2 Chemical and physical analysis. The chemical and physical analysis of the material shall be as specified in Table I.

Table I. Chemical and physical analysis.

| Characteristic | Minimum | Maximum |
|---|---|---|
| Distillation temperature, degrees C at 760 milliliters barometric pressure | | |
| For 5 milliliters of distillate | 310 | --- |
| For 50 milliliters of distillate | 320 | --- |
| Chlorides, percent | --- | 0.5 |
| Moisture, percent | --- | 1.0 |

3.3 Workmanship. The tetraethylenepentamine shall be uniform in

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quality, free from foreign materials, and manufactured under controlled conditions and procedures standard in the industry.

3.4 Warning. Tetraethylenepentamine is a hazardous liquid and vapor may cause skin irritation.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Acceptance inspection. Acceptance inspection shall, at the option of the Government, be performed either by the supplier or by a Government representative. In either case, responsibility for final acceptance rests with the Government representative.

4.2 Preproduction sample. The preproduction sample shall be prepared using the same methods proposed for the preparation of subsequent production lots of tetraethylenepentamine. The preproduction sample shall be subjected to all examinations and tests specified herein. Unless otherwise specified, the Government will perform the examinations and tests for preproduction sample acceptance at the contractor's plant. Preproduction samples which do not meet all the requirements of this specification shall be rejected and returned to the contractor. Subsequent quantities will not be considered for acceptance until approval of the preproduction sample has been obtained.

4.3 Lot. A lot shall consist of material produced at one plant with no change in formulation or process. If manufacture is by batch process, each batch shall constitute a lot. A batch shall be defined as given in 6.3.

4.4 Acceptance sampling. The number of containers to be chosen at random for acceptance sampling shall be equal to the square root of the total number of containers in the lot. If the number thus obtained is not a whole number, the number of containers to be sampled shall be increased to the next higher whole number. In no case, however, shall the number of containers to be sampled be less than seven (unless there are less than seven containers in the lot, in which case each container shall be sampled).

4.4.1 Primary sample. From each selected container, a sample shall be taken from three or more places throughout the container. The total weight of the samples taken from each container shall weigh at least 50 grams (g). Each sample thus taken shall be mixed thoroughly, placed in a clean, dry container, and labeled to identify the material name, original container designation, contract number, and lot number.



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4.4.2 Composite sample. Each primary sample shall be subdivided to prepare a composite sample (not in excess of 500 g). Primary material not used shall be returned to the primary sample container. After mixing the composite sample throughly, the composite sample shall be placed in a clean, dry container and sealed. The composite sample shall be identified with the material name, container designation, contract number, and lot number. All specified chemical and physical tests shall be made on this composite sample representing the lot. Failure of the composite sample to pass all of the tests herein shall reject the lot represented.

4.5 Classification of tests. Inspection and testing of the tetraethylenepentamine shall be classified as follows:

a. Preproduction tests.

b. Quality conformance tests.

4.6 Preproduction tests. Preproduction tests shall be conducted only on the preproduction sample and shall consist of all the examinations and tests specified herein.

4.7 Quality conformance tests. Quality conformance tests for acceptance of the tetraethylenepentamine shall consist of the following examinations and tests:

a. Distillation temperature, See 4.8.1.

b. Chlorides, see 4.8.2.

c. Moisture, see 4.8.3.

4.8 Tests. The following procedures shall be used to determine that the requirements of this specification have been met. Any changes in the test procedures employed shall be subject to prior approval of the procuring activity. Unless otherwise specified, all tests shall be run in duplicate. The average of the two results shall be taken as the test result.

4.8.1 Distillation temperature. The method of distillation is adapted from ASTM D 86.

4.8.1.1 Apparatus. Complete details and specifications for the apparatus are given in ASTM D 86. The more important items, together with the equipment difference from that specified in ASTM D 86, are described as follows:

a. Standard 200 milliliter (ml) benzol distilling flask, conforming to the following approximate dimensions: bulk, 76 millimeters (mm) outside diameter; neck, 105 mm in length by 23 mm outside diameter; vapor tube, 127 mm in length, sealed at a 75-degree angle to the neck of the flask at a point 50 mm from the top of the flask.

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b. Thermometers of suitable range, calibrated for 100-mm immersion, graduated in 0.2 degree Celsius (C) (0.36 degree Fahrenheit (F)) and capable of being easily read to 0.1 degree C (0.18 degree F).

c. Standard distillation shield and condenser. A multiple-unit system may be used if convenient.

d. Hard asbestos board, fitted tightly inside the shield and having a 4-inch circular opening.

e. Hard asbestos board, 6 by 6 by 0.25 inches, having a circular opening 1.5 inches in diameter.

f. Pittsburgh-Universal gas burner or equivalent with flame retainer, having a burning tube 155 mm long and 11 mm in diameter and clamped to a suitable ring stand so that the top of the burner is 55-60 mm below the bottom of the flask.

g. Standard 100-milliliter (ml) cylinder, graduated in milliliters, with markings numbered every 10 ml from the bottom.

4.8.1.1.1 Preparation of apparatus.

a. Assemble the apparatus as shown in figure 2 of ASTM D 86.

b. Fit the thermometer and side arm of the flask with suitable stoppers.

c. Fill the condenser with water and maintain the level so that the condenser tube is completely covered during the distillation.

d. Fasten a piece of soft lint-free cloth to a piece of cord or copper wire and swab the inside of the condenser tube to remove any liquid remaining form the previous test.

e. Measure 100 ml of the sample in the graduate to be used as the receiver. Transfer the contents of the graduate to the distillation flask, taking care not to permit any liquid to flow into the vapor tube.



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This is best accomplished by holding the flask so that the vapor tube extends in a vertical position. In the case of viscous liquids, allow the graduate to drain sufficiently.

f. Place the charged flask in the 1.5-inch opening of the 6-by-6-inch asbestos board, fitting the vapor tube into the condenser tube by means of the stopper.

g. Add a small piece of boiling stone to prevent sudden etullition of the sample during distillation.

h. Insert the thermometer in the neck of the flask by means of the stopper so that the lower end of the capillary tube is on the level with the inside of the bottom of the vapor tube at its junction with the neck of the flask. For thermometers having an expansion chamber, position the thermometer in the neck of the flask with the expansion bulb just below the bottom edge of the vapor tube junction.

i. Place the graduate used to measure the charge, without drying, at the outlet of the condenser tube in such a position that the tube extends into the graduate at least 25 mm, but not below the 100-ml mark.

j. Position the burner so that the tube is centered directly below the 1.5-inch opening in the asbestos board. When the burner is lit, the flame should burn with a hissing but not roaring sound. The distinct inner blue cone, the apex of which is intersected about 3 to 4 mm by the bottom of the flask, should halo around the opening in the asbestos board.

4.8.1.2 Distillation procedure.

a. When everything is in readiness, light the burner and apply heat uniformly so that the first drop of condensate falls from the condenser in not less than 10 nor more than 15 minutes.

b. Read and record the temperature on the thermometer when the first drop falls from the condenser as the initial boiling point. Immediately move the graduate so that the end of the condenser tube touches the inside wall of the graduate.



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c. Record the barometric pressure in mm of mercury.

d. Regulate the burner so that the distillation proceeds at a rate of 4 to 5 ml per minute.

e. Read and record the temperature corresponding to the following volumes of distillate: 5 ml plus or minus 0.5 ml and 50 ml plus or minus 0.5 ml.

f. If at any time during the distillation the sample decomposes, as evidenced by excessive fuming and a drop in the vapor temperature, stop the distillation and record the highest temperature reached as the "decomposition point". Also record the total volume of distillate collected.

g. Correct the temperature reading at 5 ml and 50 ml of distillate for the thermometer calibration constant:

$$t_c \text{ equal } t_o \text{ plus } C$$

Where: $t_o$ equals observed temperature, degrees C

$t_c$ equals temperature corrected for calibration, degrees C

C equals thermometer correction, degrees C

h. Apply the correction for barometric pressure to the corrected temperature in 4.8.1.2g to obtain the distillation temperatures for a volume of 5 ml and 50 ml of distillate.

Distillation temperature, degrees C, equals $t_c$ plus 0.063 (760 minus P)

Where: $t_c$ equals temperature corrected for calibration, degrees C

0.063 equals correction factor for every millimeter the barometric pressure is under 760 mm, degrees C

760 equals standard barometric pressure, mm

P equals actual barometric pressure during test, mm

4.8.1.3 Acceptance criteria. For the lot represented to pass the distillation temperature test, the values obtained for the distillation temperature at 5 ml and 50 ml of distillate shall be not less than specified in 3.2.



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4.8.2 Chlorides. Introduce 10 milliliter (ml) of the sample into a 40-ml porcelain crucible by means of a transfer pipet. Evaporate the sample to dryness by heating slowly with a Fisher burner. (Take care to prevent lose of the sample by spattering.) Ignite the crucible to a dull red heat over a Fisher burner for 10 minutes, and allow to cool to room temperature. Transfer the crucible to one of two 600-ml beakers and add sufficient distilled water to cover it. To the other beaker add an equivalent amount of distilled water and reserve as the blank. Place the beakers on a 500-watt electric heater, heat the solutions to boiling, and allow to boil gently for 5 minutes. Remove the source of heat and allow the solutions to cool to room temperature. To each beaker add 6 to 8 drops of 1.0-percent alcoholic phenolphthalein indicator and 2 normal (N) nitric acid dropwise until the pink color just disappears. Add one ml in excess. Slowly add powdered commercially pure calcium carbonate until effervesence ceases. Pipet exactly 5 ml of standard 0.1N sodium chloride into each beaker. Add 5 ml of a 2.0-percent aqueous dextrin solution and 10 drops of 0.1-percent alcoholic dichlorfluorescein indicator. Stir the contents of the beaker vigorously with a glass rod and titrate with standard 0.1N silver nitrate to a salmon-pink endpoint. A blank using identical quantities of reagents is run simultaneously with the sample.

$$\text{Percent chlorides, by weight as sodium chloride} \quad \text{equals} \quad \frac{(A - B)\ N \times 5.85}{10 \times C}$$

Where: A equals volume of 0.1N silver nitrate used to titrate the sample, ml

B equals volume of 0.1N silver nitrate used to titrate the blank, ml

C equals the specific gravity of the sample

10 equals volume of the samples, ml

4.8.2.1 Acceptance criteria. For the lot represented to pass the chlorides test, the value obtained for the percent chlorides as sodium chloride shall be not greater than the value specified in 3.2.

4.8.3 Moisture.

4.8.3.1 Apparatus. The apparatus used for determination of moisture content of the sample shall be an Aquameter, Model KF-2 or KF-3, Beckman Instruments, Inc., Fullerton, California, or an approved equivalent. The Aquameter shall be prepared for operation as described in the technical manual furnished by the manufacturer (Beckman Instruments, Inc.). Use of an alternate equivalent item of equipment approved by the procuring activity will necessitate use of the specific technical manual prepared by the manufacturer.



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4.8.3.2 Reagents.

a. Karl Fischer reagent. Karl Fischer reagent must have a strength such that each milliliter of Karl Fischer reagent corresponds to 0.0014-0.0023 g of water. Dilute 750 ml of commercially available stabilized Karl Fischer reagent (with water equivalent of 0.005-0.007 g/ml) to 2000 ml with absolute methanol (0.1 percent water, maximum). Mix well and allow to stand overnight before use. Determine the water equivalent (A) of this solution as follows:

1. Use sodium tertrate dihydrate ($Na_2C_4O_6$ $2H_2O$) a primary standard (with a water content of 15.66 percent) for standardizing Karl Fischer reagent. If the water content value is in question, it may be determined by heating some of the salt at 150 degrees C (302 degrees F) for 3 hours. Should the value (as determined) differ from the theoretical value of 15.66, then the experimental value shall be used in the determination of water equivalent (A) of the Karl Fischer reagent; instead of the 15.66 in the formula below, the factor should be 10P where P is percentage moisture (as determined). Rapidly transfer 0.090-0.110 g (weighed to the nearest 0.0001 g) of reagent-grade ($Na_2C_4H_4O_6$ $2H_2O$) to the titration vessel.

2. Titrate to an endpoint in the same manner as with the sample. (See 4.8.3.3.)

3. Repeat the standardization procedure until three successive results agree within five parts per thousand.

4. If the indicated water equivalent (A) of the Karl Fischer reagent is less than 0.0014 g of water per milliliter of Karl Fischer reagent, it may be due to the presence of too much water in the absolute methanol used. In this case, distill the methanol from metallic calcium or calcium hydride. Passing the methanol through a column of Molecular Sieves, Type 4A, may also reduce the water content of the methanol sufficiently.



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Water equivalent (A) equals $\frac{0.1566W}{V}$

Where: A equals water equivalent of the Karl Fischer reagent, g/ml
W equals weight of $N_aC_4H_4O_6\ 2H_2O$
V equals volume of Karl Fischer reagent used, ml

b. Water-in-methanol solution. The water-in-methanol solution should contain 0.0015-0.0020 g of water per milliliter of solution. A good grade of commercial absolute methanol contains about 0.0010 g water per milliliter of methanol. Water content can be adjusted by adding 1.0 g of water to 1000 ml of the water-in-methanol solution to produce a change of 0.0010 g per milliliter. Determine the relative strength of the water-in-methanol solution in terms of Karl Fischer reagent as follows:

1. Put about 50 ml of the anhydrous methanol used in 4.8.3.2a into the titration beaker of the Aquameter. Add a slight excess of Karl Fischer reagent (4.8.3.2a), then back-titrate with water-in-methanol solution (4.8.3.2b). Then run in additional 5 to 8 ml of Karl Fischer reagent, read to the nearest 0.01 ml, and again back-titrate with water-in-methanol solution (read to the nearest 0.01 ml). Repeat the addition and back-titrating steps twice more to provide triplicate determinations of the equivalency ratio. Calculate the ratio (B) of Karl Fischer reagent to that of the water-in-methanol solution. The range of the ratios calculated from the three titrations should be not greater than 0.04. If the range exceeds 0.04, continue making titrations until three ratios are obtained whose range does not exceed 0.04. Then determine the average ratio from all the ratios which have been obtained.

4.8.3.3 Procedure. Determine the moisture content of the sample by the Karl Fischer method using a direct-titration technique. Introduce to the titration beaker, through the opening in the diaphragm, approximately 10 g of sample weighed to the nearest 0.001 g and a mixture of 150 ml of anhydrous methanol with 50 ml of glacial acetic acid. Both materials are to be reagent grade. Close the opening, start the stirrer, and press the titrate button to titrate with Karl Fischer reagent (4.8.3.2a). When the indicator light glows, read the Karl Fischer buret to the nearest 0.01 ml. Where necessary, water-in- methanol solution (4.8.3.2b) may be used to back-titrate.



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$$\text{Percent moisture equals } \frac{100A\ (V_{KF} \text{ minus } BV_{WM})}{W}$$

Where: A equals weight of water equivalent to 1.00 ml of Karl Fischer reagent, g/ml

$V_{KF}$ equals volume of Karl Fischer reagent titrant used, ml

$V_{WM}$ equals volume of water-in-methanol solution titrant used, ml

B equals ratio of Karl Fischer reagent to that of water-in-methanol solution, ml/ml

W equals weight of sample taken, g

4.8.3.4 Acceptance criteria. For the lot represented to pass the moisture test, the value obtained for the percent moisture shall be not greater than the value specified in 3.2.

5. PREPARATION FOR DELIVERY

5.1 Preservation, Packaging, and Packing. Unless otherwise specified by the contract or purchase order, preservation, packaging, and packing requirements for the specified levels of protection shall be in accordance with MIL-P-14232.

5.2. Marking. In addition to the markings required by contract or order, unit packages and shipping containers shall be marked and labeled in accordance with the requirements of MIL-STD-129. If the material has toxic characteristics, the proper toxicity precautions shall be listed on the package.

6. NOTES

6.1 Intended use. Tetraethylenepentamine is intended for use as a catalyst in an epoxy potting compound for humidity and pressure seals.

6.2 Ordering data. Procurement documents should specify the following information:

a. Title, number, and date of specification.

b. Whether a preproduction sample is required (see 3.1).

c. Type and size of shipping container (see 5.1).



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6.3 Definition.

6.3.1 Batch. A batch is defined as that quantity of material which has been subjected to one or more chemical or physical processes (or combinations thereof) intended to produce a desired product having substantially uniform characteristics.

6. Supersession data. This specification includes the requirements of Missile Interim Specification MIS 14164A, dated, 23 December 1965.

Custodian:
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Preparing Activity:
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